DSP Engine: Audio Analysis Core

FFT, LUFS, Correlation, Clipping and Real-Time Metrics

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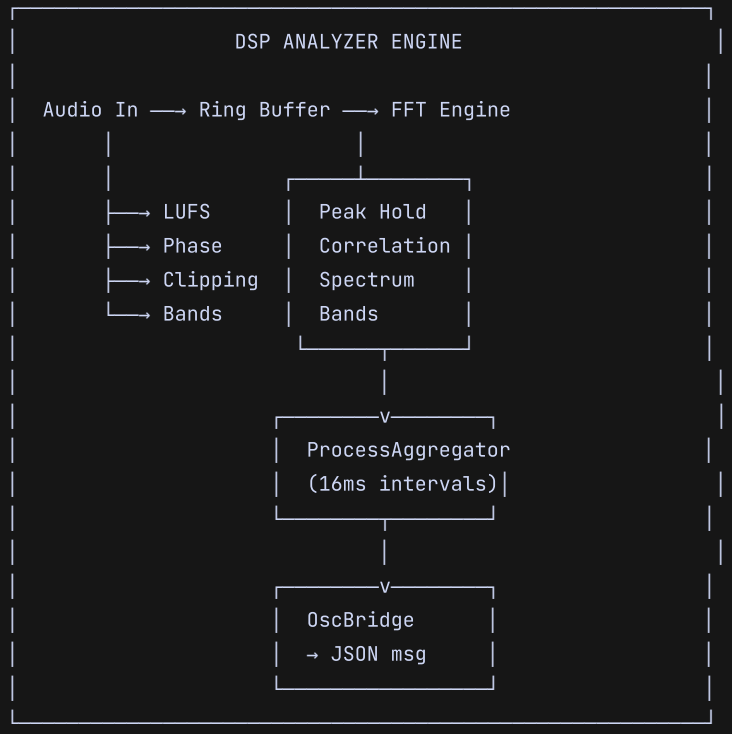
Category: Audio Processing / DSP
Dependencies: JUCE DSP Module, FFTReal, SIMD (optional)

1. DSP Engine Overview

The WhyCremisi DSP analysis engine is the AI agent's sensory system — its equivalent of the human auditory system. It operates on real-time audio buffers (64–1024 samples) and produces structured metrics that feed both the UI and the AI decision models.



Every processed audio block passes through the full analysis chain. Results are aggregated every 16ms (approximately 62 times per second at 48kHz) and sent to the presentation layer.

2. Analyzer Architecture

2.1 FFT ANALYSIS CHAIN

Spectral analysis is based on a Hanning-windowed FFT:

```
FFT Parameters:
──────────────────────────────────────────────
Window size           2048 samples
Hop size              512 samples       (75% overlap)
Window                Hanning (von Hann)
Zero-padding          No
Minimum frequency     21.5 Hz  (at 48kHz)
Maximum frequency     24.0 kHz  (Nyquist)
Frequency resolution  23.4 Hz  (per bin at 48kHz)
──────────────────────────────────────────────
```

[NOTE] The 75% overlap guarantees a temporal resolution of ~10.6ms at 48kHz, sufficient for perceptually relevant transients while keeping computational load under control.

```
FFT Implementation (pseudocode):

void Analyzer::processFFT(const float* channelData, int numSamples) {
    // Copy into FFT circular buffer
    fftBuffer.write(channelData, numSamples);

    if (fftBuffer.available() ≥ fftSize) {
        fftBuffer.read(windowBuffer, fftSize);

        // Apply Hanning window
        for (int i = 0; i < fftSize; ++i)
            windowBuffer[i] *= hanningWindow[i];

        // Execute FFT
        fft.performRealOnlyForwardTransform(windowBuffer);

        // Compute magnitude (dB) per bin
        for (int bin = 0; bin < numBins; ++bin) {
            float real = windowBuffer[bin * 2];
            float imag = windowBuffer[bin * 2 + 1];
            magnitude[bin] = sqrtf(real * real + imag * imag);
            magnitudeDB[bin] = 20.0f * log10f(magnitude[bin] + 1e-12f);
        }
    }
}
```

2.2 SPECTRUM BINS

With FFT size 2048, the number of usable bins (up to Nyquist) is 1024. Each bin has a width of ~23.4Hz at 48kHz.

BIN	FREQUENCY (Hz)	NOTE
0	0	DC component (discarded)
1	23.4	Extreme sub bass
2	46.9	Sub bass
...	...	
43	1007.8	Low-mids
128	3000.0	Presence
256	6000.0	High-mid
512	12000.0	Brilliance
1024	24000.0	Nyquist (discarded)

2.3 STEREO CHANNELS

Analysis is performed independently on both channels:

```
struct StereoAnalysis {
    float leftMagnitude[1024];
    float rightMagnitude[1024];
    float leftPhase[1024];
    float rightPhase[1024];
    float correlationValue;    // [-1.0, 1.0]
    float midMagnitude[1024];  // L+R
    float sideMagnitude[1024]; // L-R
};
```

3. Loudness Metrics

3.1 LUFS ALGORITHM (ITU-R BS.1770-4)

WhyCremisi implements loudness measurement per the ITU-R BS.1770-4 standard, with three time windows:

Type	Window	Update Rate
Momentary (M)	400 ms	Every 100 ms
Short-term (S)	3 s	Every 300 ms
Integrated (I)	∞ (cumul.)	Every block

— 3.2 GATING

ITU-R BS.1770-4 requires a two-stage gate for Integrated LUFS calculation:

```
Gating Algorithm:

Stage 1 — Absolute Gate:
 Threshold: -70 LUFS (silence)
 Blocks below threshold → excluded from calculation

Stage 2 — Relative Gate:
 Compute average loudness of blocks above -70 LUFS
 Relative threshold: average + 10 LUFS (typical: -25 LUFS)
 Blocks below relative threshold → excluded from calculation

Result: Integrated LUFS = average of remaining blocks
```

[NOTE] Relative gating is crucial: without it, silence between tracks would artificially lower the integrated loudness value, leading to non-compliant broadcast masters.

— 3.3 TRUE PEAK

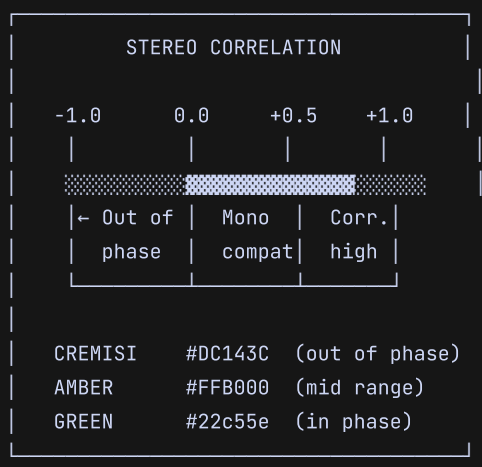
True Peak is calculated via 4x oversampling with sinc reconstruction filter:

```
True Peak = max(|signal oversampled 4x|)

Precision: ±0.1 dB (ITU-R BS.1770-4 compliant)
Used for: safe limiting, broadcast mastering
```

4. Phase Correlation

4.1 PHASE CORRELATION METER

The correlation meter measures similarity between left and right channels:



Mathematical formula:

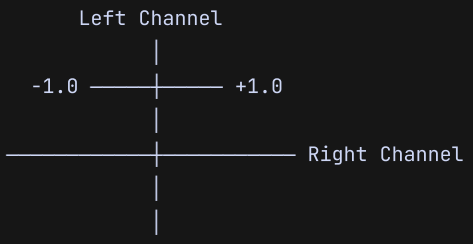
```
correlation = Σ(L[n] * R[n]) / sqrt(Σ(L[n]²) * Σ(R[n]²))

Interpreted values:
─────────────────────────────────────────────
+1.0  →  Identical (perfect mono)
+0.5  →  Good mono compatibility
 0.0  →  Completely decorrelated
-0.5  →  Evident phase problems
-1.0  →  Total phase inversion
─────────────────────────────────────────────
```

4.2 MID/SIDE ANALYSIS

```
Mid  (M) = L + R    (everything common)
Side (S) = L - R    (everything different)

M/S Ratio:
─────────────────────────────────────────────
M >> S   →  Narrow, centered mix
M = S    →  Balanced, spacious mix
M << S   →  Wide mix, potential phase issues
─────────────────────────────────────────────
```

4.3 VECTORSCOPE / PHASE SCOPE

The vectorscope is a 2D L/R display tracking instantaneous amplitude:

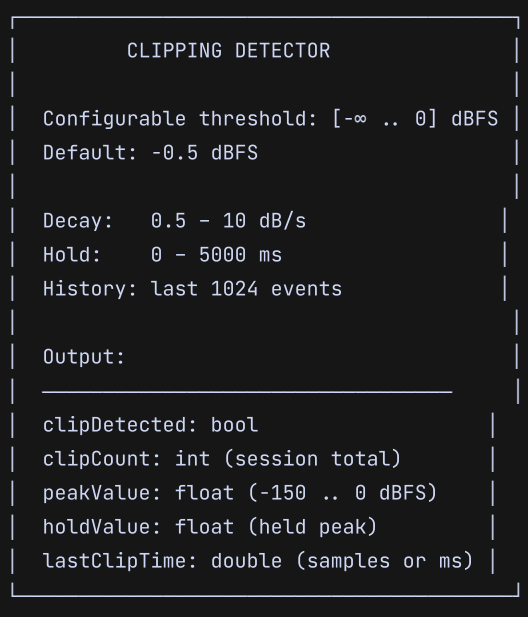


Characteristic shapes:

45° line	→	Perfect mono
Circle	→	Balanced stereo
Horizontal line	→	Left only (panned L)
Vertical line	→	180° out of phase

5. Clipping Detection

5.1 PEAK HOLD WITH RMS



5.2 HISTORY BUFFER

```
struct ClipEvent {
    double timestamp;    // absolute sample position
    int    trackIndex;   // track ID
    int    channelIndex; // 0=L, 1=R
    float  peakValue;    // peak value in dBFS
};

class ClipHistory {
    std::array<ClipEvent, 1024> buffer;
    int writeIndex;
    int count;
    std::atomic<int> totalClips;

    void recordClip(ClipEvent event);
    const ClipEvent* getLastClip() const;
    int  getTotalClipCount() const;
    void autoReset(double timeoutMs);
    void clear();
};
```

5.3 AUTO-RESET

```
Reset modes:
─────────────────────────────────────────────
Manual          → User clicks "reset"
Timeout         → After N ms without new clips
Threshold       → Clip below new threshold
Track change    → Current track changes
─────────────────────────────────────────────
```

6. Frequency Spectrum

6.1 LOG SCALE 20HZ–20KHZ

The frequency axis is mapped on a logarithmic scale to match human perception:

```
Pixel → frequency mapping (768px display):

px      freq (Hz)   note
──────────────────────────
0       20          Sub bass
128     55          A1 (bass)
256     151         D3
384     415         G#4
512     1140        D#6
640     3136        G7
768     20000       Brilliance
──────────────────────────

Formula: freq = 20 * pow(20000/20, pixel / width)
```

6.2 AGGREGATE FREQUENCY BANDS

```
Band           Range         FFT Bins   UI Color
──────────────────────────────────────────────────
Sub            20 – 60 Hz    0 – 2      #DC143C  (cremisi)
Bass           60 – 250 Hz   2 – 10     #FF6B6B
Low-Mid        250 – 500 Hz  10 – 21    #FFB000  (amber)
Mid            500 – 2 kHz   21 – 85    #FFD700
High-Mid       2 – 6 kHz     85 – 256   #22c55e  (green)
Presence       6 – 10 kHz    256 – 426  #00E5FF  (cyan AI)
Brilliance     10 – 20 kHz   426 – 853  #00FFaa  (OK green)
──────────────────────────────────────────────────
```

```
struct FrequencyBand {
    const char* name;
    float freqMin, freqMax;
    int   binStart, binEnd;
    float energy;        // sum of magnitudes in bin range
    float peak;          // peak energy in range
    int   peakBin;       // peak bin index
    float energyDB;      // energy in dB
};

void Analyzer::computeBands(const float* magnitude,
                            FrequencyBand* bandsOut) {
    for (int b = 0; b < NUM_BANDS; ++b) {
        auto& band = bandsOut[b];
        float sum = 0.0f;
        float maxVal = -INFINITY;
        int maxBin = 0;

        for (int bin = band.binStart; bin < band.binEnd; ++bin) {
            sum += magnitude[bin];
            if (magnitude[bin] > maxVal) {
                maxVal = magnitude[bin];
                maxBin = bin;
            }
        }

        band.energy = sum;
        band.peak = maxVal;
        band.peakBin = maxBin;
        band.energyDB = 20.0f * log10f(sum + 1e-12f);
    }
}
```

— 6.3 SPECTRAL PEAK HOLD

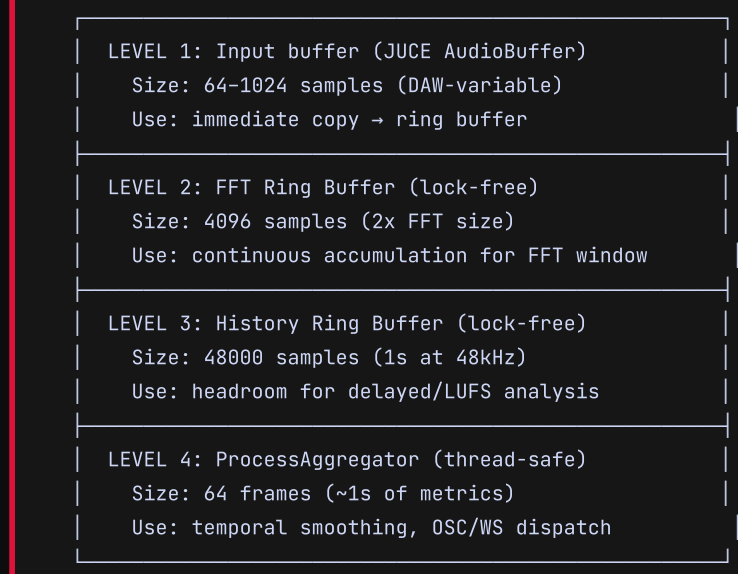
```
Peak Hold:
──────────────────────────────────────────────
Decay rate:     2 dB/s  (adjustable 0.5–10)
Initial hold:   500 ms  (no decay on fresh peak)
Update:         every FFT frame
Reset:          on request or track change

Display:
─── Instantaneous spectral curve (fill)
─── Spectral peak hold curve (thin line)
```

[NOTE] Slow-decay peak hold is essential for rapid identification of problem frequencies: a momentary peak is kept visible long enough for the AI to suggest a targeted EQ intervention.

7. Performance and Optimization

7.1 BUFFER MANAGEMENT

Buffer hierarchy:



7.2 LOCK-FREE RING BUFFER

```
template<typename T, size_t N>
class LockFreeRingBuffer {
    std::array<T, N> buffer;
    std::atomic<size_t> writePos{0};
    std::atomic<size_t> readPos{0};

public:
    bool write(const T* data, size_t count) {
        size_t wp = writePos.load(std::memory_order_relaxed);
        for (size_t i = 0; i < count; ++i) {
            buffer[(wp + i) % N] = data[i];
        }
        writePos.store((wp + count) % N,
                       std::memory_order_release);
        return true;
    }

    bool read(T* out, size_t count) {
        size_t rp = readPos.load(std::memory_order_relaxed);
        // ... mirror implementation
        return true;
    }

    size_t available() const {
        size_t wp = writePos.load(std::memory_order_acquire);
        size_t rp = readPos.load(std::memory_order_acquire);
        return (wp ≥ rp) ? (wp - rp) : (N - rp + wp);
    }
};
```

— 7.3 SIMD CONSIDERATIONS

```
Vectorizable operations (with SIMD/AVX):
──────────────────────────────────────────────────
✓ Hanning window multiplication (vector × vector)
✓ Magnitude: sqrtf(real² + imag²) for 8 bins
✓ LUFS sum: mean square over 8-sample blocks
✓ Correlation: vectorized L[n] × R[n] product

Estimated speedup vs. scalar:
──────────────────────────────────────────────────
Hanning window      ~4×
Magnitude compute   ~6×
LUFS mean square    ~4×
Correlation         ~3×
──────────────────────────────────────────────────
```

— 7.4 THREAD SAFETY

```
Concurrency model:

AUDIO THREAD (real-time, highest priority)
├── FFT ring buffer write
├── History ring buffer write
├── FFT execution ✓
├── LUFS momentary computation ✓
└── Clipping detection ✓
└── ❌ No heap allocations
    ❌ No mutex locks
    ❌ No I/O (file/socket)

UI THREAD (main thread)
├── Aggregated result reads
├── Graphical rendering
└── OSC/WebSocket message dispatch

AGGREGATOR THREAD (16ms timer)
├── Data reads from atomic buffers
├── Derived metric computation
├── Temporal smoothing
└── JSON message preparation
```

[NOTE] Strict separation between audio and UI threads is critical: the audio thread must never be blocked by I/O operations or memory allocations. All communication occurs through lock-free atomic data structures.

8. System Integration

8.1 PLUGINPROCESSOR CONNECTION

```
PluginProcessor (JUCE)
     │
     │  processBlock(audioBuffer, midiMessages)
     ▼
Analyzer::processBlock(const float** channelData,
                       int numChannels, int numSamples)
     │
     ├── per channel:
     │     ├── processFFT(channel, numSamples)
     │     ├── processLUFS(channel, numSamples)
     │     ├── processCorrelation(channel, numSamples)
     │     └── processClipping(channel, numSamples)
     │
     └── Analyzer::aggregate()
           └── computeBands(magnitude, bands)
           └── computeMeterData()
```

8.2 OSCBRIDGE DISPATCH

Aggregated data is serialized to JSON and sent via WebSocket:

```
{
  "type": "daw.analyzer",
  "timestamp": 1715123456789,
  "sampleRate": 48000,
  "metrics": {
    "loudness": {
      "momentary": -14.2,
      "shortTerm": -13.8,
      "integrated": -12.5,
      "truePeak": -1.2
    },
    "correlation": 0.72,
    "clipping": {
      "detected": false,
      "totalClips": 0,
      "peak": -6.3
    },
    "spectrum": {
      "bands": [
        { "name": "sub", "energy": -28.4, "peak": -22.1 },
        { "name": "bass", "energy": -18.2, "peak": -14.5 },
        { "name": "lowMid", "energy": -22.7, "peak": -18.3 },
        { "name": "mid", "energy": -16.5, "peak": -12.2 },
        { "name": "highMid", "energy": -24.1, "peak": -19.8 },
        { "name": "presence", "energy": -30.6, "peak": -26.4 },
        { "name": "brilliance", "energy": -35.2, "peak": -31.0 }
      ]
    }
  }
}
```

8.3 UPDATE FREQUENCIES

Metric	Frequency	Max Latency
LUFS Momentary	Every 100 ms	100 ms
LUFS Short-term	Every 300 ms	300 ms
LUFS Integrated	Every block	N/A (cumul.)
True Peak	Every block	5.3 ms
Correlation	Every 16 ms	16 ms
Clipping	Every 16 ms	5.3 ms (1 hop)
Spectrum bands	Every 32 ms	32 ms
Full FFT spectrum	Every 16 ms	16 ms (full frame)

9. Limitations and Future Improvements

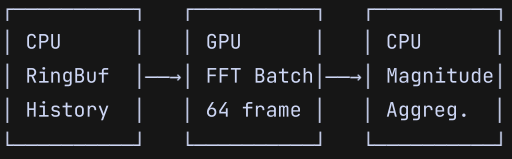
9.1 CURRENT LIMITATIONS

Limitation	Impact
CPU-only FFT	High CPU load at high sample rates
Mono/Stereo only	No multichannel support (5.1, 7.1)
No source separation	Rhythm overlaps mask spectral analysis
Fixed Hanning window	Improvable with adaptive windows
No ambient compensation	Background noise unfiltered

9.2 PLANNED IMPROVEMENTS

GPU Acceleration for FFT

Using CUDA/Metal Performance Shaders for concurrent FFT across multiple channels:



Estimated speedup: 10–50× for batch FFT ops
Target: Integrated GPU (Apple Silicon, Intel Iris)

Machine Learning for Source Separation

```
Pre-processing with lightweight model (reduced
Demucs or Spleeter) to separate signal into
components (voice, drums, bass, other) before
analysis:

Audio In → Source Separation → Analyzer per
                                source
                           → EQ suggestion per
                             specific source
```

Multichannel Expansion

Incremental Enhancements

- Adaptive windows (Kaiser with variable beta for resolution/spectral leakage trade-off)
- Optional zero-padding for high-resolution FFT
- Automatic noise floor detection with adaptive thresholding
- Real-time sonogram with perceptual compression (optional mel scale)
- Spectrum cache for instant A/B comparison between plugin states

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References

1. ITU-R BS.1770-4 — Algorithms to measure audio programme loudness and true-peak audio level (2015)
2. FFTPack / JUCE DSP Module — Reference FFT implementation
3. Zölzer, U. — Digital Audio Signal Processing (3rd ed., Wiley 2022)
4. Smith, J.O. — Spectral Audio Signal Processing (online, 2011)
